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PRISM: making glioma progression scores inspectable on post-treatment MRI

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ABSTRACT

Following treatment for aggressive glioma, changes on serial MRI must be interpreted alongside the treatment course. A progression score can rank surveillance scans, but cannot show whether it reflects MRI measurements or treatment context. We developed PRISM, a local system that makes progression scores inspectable by combining tumour measurements and treatment context in a checked report. On MU-Glioma-Post, the development cohort, PRISM distinguished scans with recorded progression from other scans (AUC 0.82). Applied without refitting to external cohorts, prespecified models reached AUCs of 0.77 at LUMIERE (combined) and 0.80 at UCSF-ALPTDG (single scan images). The scores were linked to treatment timing at MU and enhancing volume at UCSF. PRISM shows each score with the measurements, treatment context and provenance behind it. After one revision of the prompt, all 406 reports across the three cohorts met predefined automatic checks against their structured records. Future work will evaluate the reports with clinicians.

KEYWORDS

glioma surveillance, external validation, conformal prediction

1. Introduction

After surgery and chemoradiotherapy for glioblastoma and other aggressive gliomas, serial MRI is read against earlier scans and the course of treatment. One early challenge is pseudoprogression. Treatment can produce new or increasing enhancement that resembles tumour growth. Pseudoprogression is concentrated within 12 weeks of completing chemoradiotherapy (Wen et al. 2023) and is more common in tumours with MGMT promoter methylation (Taal et al. 2008; Brandes et al. 2008).

Models intended to predict progression assign a score to each surveillance scan. On its own, a score gives an assessment without showing the information that informed it. It does not show whether the available information agrees, whether relevant information is missing, or how the model has performed in comparable clinical situations.

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PRISM is a system that makes progression scores inspectable. For each scan, it keeps image measurements, treatment context, and separate model scores together in a record that is checked before it is written as a report. The name reflects this design. As a prism separates light, PRISM keeps image measurements and treatment context separate. Earlier systems have generated brain tumour reports from structured features extracted from segmentation (Heras Rivera et al. 2026; Valerio et al. 2025), and deterministic checks have compared generated clinical text with source records (Zeba et al. 2026). PRISM applies these ideas to glioma surveillance after treatment. Its language model writes the completed record as text, and an automatic checker compares the text with that record.

We developed PRISM on the public MU-Glioma-Post cohort and validated it, without refitting, in the LUMIERE and UCSF-ALPTDG cohorts. This paper makes four contributions. The first two are the methods we built, and the last two are what testing them showed.

(1) This paper introduces PRISM, a framework that builds one record for each follow up scan. The record holds tumour volumes and their change measured from segmentation masks, three separately fitted scores (a clinical-history model, an imaging model and a combined model), and three statements of confidence (an 80% conformal set, whether the clinical-history and imaging models agree, and how the models performed in comparable situations), each value with its source (Sections 2.1–2.3, validated in Sections 4.1 and 4.2).
(2) This paper presents a reporting workflow that runs entirely on local hardware. A language model writes the report from the record alone, citing for each sentence the record fields it uses, and a deterministic checker rejects any report that states a number, flag or confidence statement absent from the record (Section 2.4, validated in Section 4.3).
(3) This paper shows that progression scores with similar discrimination can rest on different information in different cohorts. The scores followed treatment timing at MU and enhancing volume at UCSF, which an AUC alone cannot reveal (Sections 3.4 and 4.4).
(4) This paper validates PRISM in two external cohorts with models fitted once on MU and rules fixed in advance. It tests whether the scores still rank scans, whether the confidence statements keep their stated coverage, and whether every report agrees with its record (Sections 3.2 and 4.1–4.3).

The rest of the paper follows this order. Section 2 describes PRISM, from its inputs and measurements (Section 2.1) and its three models (Section 2.2) to its confidence statements (Section 2.3), its record and report (Section 2.4), and what it costs to run (Section 2.5). Section 3 describes the three cohorts, the task, how each cohort was evaluated, which tests were fixed in advance, and the audits of what carries a score. Section 4 reports the validations in the order of the contributions, first how well the three models rank scans in each cohort and how the confidence statements behave, then the reports, and finally what information carries the scores in each cohort. Section 5 discusses what the results mean for anyone who builds or evaluates such scores, and what remains untested, above all whether the reports help clinicians.

2. PRISM

PRISM turns one follow up scan and the patient's record into a structured record and a report, in five stages (Figure 1A). It measures the tumour, scores the scan with three models, attaches three confidence statements, writes every value into one record with its source. A local language model

then writes the report from that record, and a checker compares the report with the record. Table 1 lists every score used in the paper.

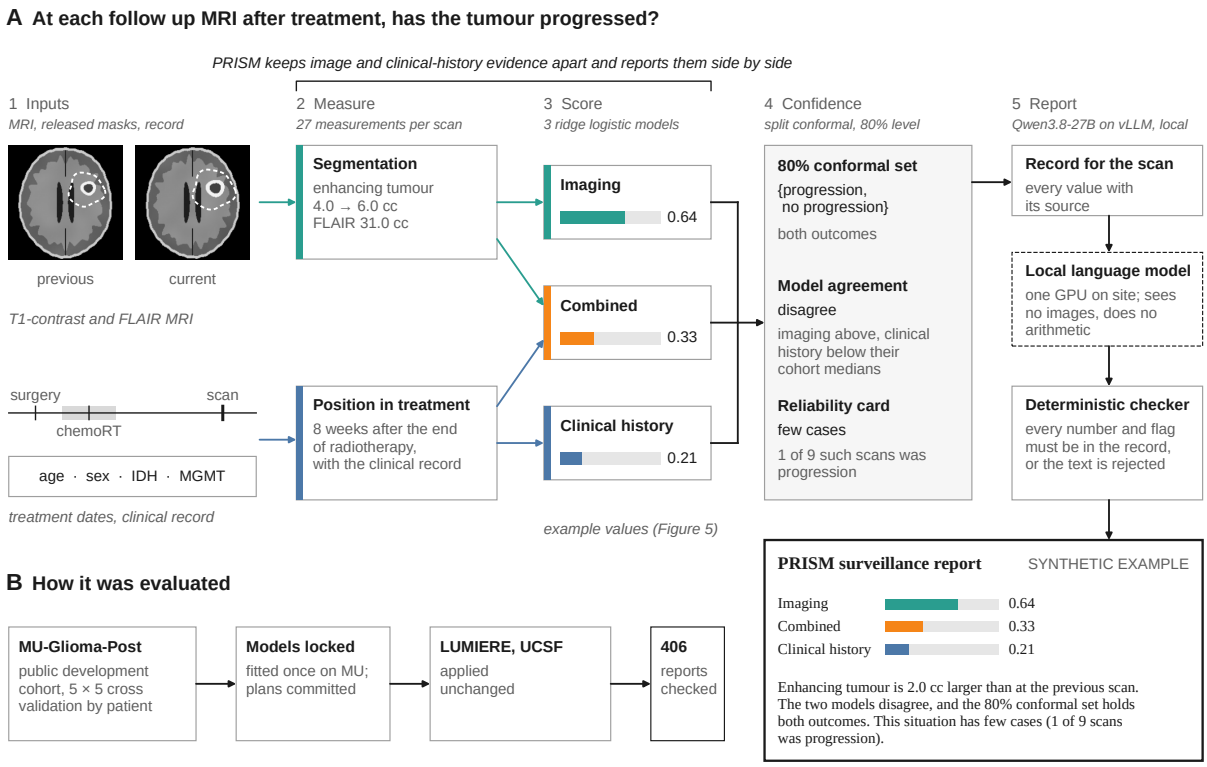


Figure 1. PRISM and how it was evaluated. (A) At each follow up MRI, PRISM keeps image and treatment evidence apart. The masks released with the cohort give the volumes of the previous and current scans and 27 measurements of the tumour and its change, which feed the imaging model. The treatment dates and clinical record give the position in treatment, which feeds the clinical-history model. A third model on both inputs gives the combined score, and all three are ridge logistic regressions (values from the synthetic example of Figure 4). Three confidence statements are attached, a split conformal set at the 80% level, the agreement of the two models and a lookup of results in the same situation. Every value enters one record with its source, a local language model (Qwen3.8-27B served by vLLM on one GPU) writes the report from the record without seeing images, and a deterministic checker tests the report against the record. (B) Models were developed on MU by repeated cross validation, then fitted once and applied unchanged at LUMIERE and UCSF under rules fixed before any score was compared with an external label.

2.1. Inputs and measurements

PRISM takes three inputs for a scan. The first is the segmentation of the current scan and of earlier scans, which gives the volumes of enhancing tumour, of FLAIR hyperintensity, and of necrosis and the resection cavity. We used the masks released with each cohort rather than segmenting again (Mahmoud et al. 2025; Suter et al. 2022; Fields et al. 2024), so every volume can be traced to a public file. From the masks and images PRISM computes 27 measurements. Nineteen describe the current scan, namely the log volumes of the six compartments (enhancing tumour, necrosis, FLAIR hyperintensity, resection cavity, tumour core and whole lesion) and 13 measures of shape and of T1 contrast and FLAIR intensity relative to normal tissue. The other eight describe change,

Table 1. The scores built from measurements and records. Every score listed is a ridge logistic regression except enhancing volume alone, which is not fitted. The others are fitted within the training data, with unknown IDH or MGMT status as the reference category and missing measurements imputed with an indicator. The networks tuned on the task are described in Section 3.4. Status indicates whether a model was planned before the analysis or not fixed in advance (†).

Score	Inputs	Dates	Status
Clinical data alone	Age, sex, IDH and MGMT status	No	Planned reference (LUMIERE †)
Clinical-history model	Clinical inputs; days since diagnosis and surgery; radiotherapy start, end and dose; chemotherapy; visit index	Yes	Planned
Reduced clinical-history model	Clinical inputs; days since surgery; flag for the first weeks after radiotherapy	Partly	Planned at UCSF-ALPTDG; † at LUMIERE
Imaging model	27 measurements of the segmented tumour: volumes, shape and intensity, change since the previous scan, volumes relative to the patient's smallest	No	† on MU; planned at UCSF-ALPTDG and in the system tests
Images from one scan	The 19 measurements of the current scan	No	† on MU; planned at UCSF-ALPTDG
Combined score	Inputs of both models; externally, the reduced clinical-history inputs, volumes and their change, and the interval since a previous scan	Yes; partly externally	† on MU (full form); planned and locked (reduced form) at LUMIERE and UCSF-ALPTDG
Planned primary model	Clinical-history inputs, image measurements and three schedule inputs (interval since the previous scan, whether there was one, smallest volume since the end of radiotherapy)	Yes	Planned primary model on MU; not part of PRISM
Enhancing volume alone	Log enhancing tumour volume, no fitted model	No	† reference

namely the log ratios of the volumes to those of the previous scan and the enhancing and lesion volumes relative to the smallest the patient has had up to that scan (Table 1). The second input is the treatment history as of the scan day, meaning the days since diagnosis and surgery and the start, end and dose of radiotherapy and chemotherapy. From it PRISM places the scan in a phase of care, before the end of radiotherapy, early after it (up to 12 weeks after its end), later, or unknown when the dates are missing. The third input is the patient's age, sex, and IDH and MGMT status. Every input is taken as of the scan day, so no information from a later visit enters a score.

2.2. Three models

PRISM scores each scan with three models. The clinical-history model uses the treatment history and the patient characteristics, the imaging model uses the 27 measurements and no dates or intervals, and the combined model uses both sets of inputs. Each is a logistic regression with an L2 (ridge) penalty, chosen because its coefficients can be read and because deep networks tuned on MU did not rank scans better than the measurements (Table 5). Missing values are imputed with the median of the training scans and flagged with an indicator, unknown IDH or MGMT status is coded as the reference category, and the inputs are standardised. The penalty is chosen from six values of the inverse strength C (0.01, 0.03, 0.1, 0.3, 1 and 3) by threefold cross validation grouped by patient within the training data, minimising the log loss. The output of each model is a score between 0 and 1 that ranks scans by how closely they resemble progression scans in the training data.

Where a cohort lacks treatment dates, PRISM uses a reduced clinical-history model with the inputs that cohort has (Table 1). The flag for the first weeks after radiotherapy is taken from the rater's note at LUMIERE and approximated at UCSF as 42 to 125 days after the radiotherapy start,

assuming a six week course. LUMIERE's masks have three compartments and no separate cavity, so for LUMIERE we fitted a separate imaging model on MU with only those three volumes and their change since the previous scan, a reduction fixed before evaluation. Because the imaging model needs only scans and masks, PRISM still gives an imaging score in a cohort with no treatment dates.

2.3. Confidence statements

PRISM attaches three confidence statements to each score and keeps them separate, so that each can be checked on its own.

The first is a conformal set at the 80% level (Angelopoulos and Bates 2023), which names the outcomes the combined score cannot rule out. We refitted the combined model on 70% of the MU training patients and used the other 30% for calibration. Write $\hat{p}(1) = \hat{p}$ for a scan's score and $\hat{p}(0) = 1-\hat{p}$. For each calibration scan i with outcome y_i, the nonconformity score is $s_i = 1-\hat{p}_i(y_i)$, the probability the model did not give to what happened. Because a patient contributes several scans that are not independent, each scan is weighted by one over the number of that patient's calibration scans, so that every patient carries the same weight. With $\alpha = 0.2$ and n calibration patients, the threshold $\hat{q}$ is the smallest s_i whose cumulative weight, with the scores sorted, reaches $(1-\alpha)(n+1)$, which counts the new patient with weight one, and $\hat{q}$ is infinite if no score reaches it. A new scan with score $\hat{p}$ receives the set

$$C(\hat{p}) = \{\, y \in \{0,1\} : 1 - \hat{p}(y) \le \hat{q} \,\}. \tag{1}$$

The set holds one outcome when the score is clear, both when it is not, and in principle none when the threshold is very small. It is computed from the model refitted on 70% of the patients, not from the displayed combined score, and this costs some discrimination (at UCSF the refitted model's AUC was 0.08 below that of the displayed score). The record shows the combined model's set. We computed sets in the same way for the clinical-history and imaging models to test them. If a new patient is exchangeable with the calibration patients, the set contains the true outcome of about 80% of their scans on average, but not necessarily within each subgroup (Romano et al. 2020). On MU the calibration split was made inside each training fold. Externally we used one model and one threshold from MU, unchanged.

The second statement records whether the clinical-history and imaging models agree. Each model's score is placed above the median of that model's scores over the cohort's evaluation scans, or otherwise below it, a rule that uses no outcome labels, and the two models agree when both scores fall on the same side. Because a cohort median needs a batch of scans, we later also tested medians fixed on MU, which can be applied to a single scan[†].

The third statement is a reliability card[†] (Mitchell et al. 2019), a lookup of how the models performed on scans in the same situation. A situation is the phase of care crossed with the state of the enhancing tumour. That state is absent when the scan has less than 0.1 cc, unknown without a usable segmentation, without comparison when there is no previous scan, increased when it grew by at least 1 cc and 40% from a measurable previous volume or by at least 1 cc of new enhancement from none, and not increased otherwise. For each cohort the card counts the scans with and without progression in that situation among the cohort's held out results. With at least 10 of each it shows

the AUC of all three models with 95% confidence intervals (tested), with fewer it shows the counts only (few cases), and with none it says the situation was not tested. Because those results include the scan itself, the card describes it slightly optimistically. PRISM also flags a scan whose IDH or MGMT status is unknown or whose inputs lie outside the 1st to 99th percentile of the training data.

2.4. Record and report

Every value PRISM computes for a scan enters one structured record under a fixed key, with its source. A local language model (Qwen3.8-27B, run with greedy decoding) then writes the report from that record alone, so patient data do not leave the hospital. The prompt is assembled from the fields present in the record, so the model is never asked about information the record lacks, and decoding is constrained to a JSON schema in which every sentence carries its section and the keys of the record fields it uses. The model sees no images and does no arithmetic, and every number it writes must match a field it cites, allowing only the rounding, sign and percentage forms the record already implies.

A deterministic checker then accepts or rejects the whole report. It rejects a report whose output cannot be parsed, in which a sentence cites a field that does not exist, states a number not found in a field it cites, or writes a number as a word, in which a raised flag is not mentioned or an input is wrongly called out of range, or in which the conformal set, the agreement statement or the reliability tier is missing or misstated. It also rejects a report that leaves out any of the three scores or the enhancing volume, that quotes a performance figure absent from the reliability card, or that offers a diagnosis, a response category, advice, certainty words such as “rules out” or a verbal probability such as “likely”. A rejected report is not repaired. The checker can confirm that a report matches its record, but it cannot judge emphasis or what a clinician would infer (Supplementary Section S19).

No public cohort releases the clinicians’ reports for these scans, so we compared the report’s content with an established standard instead, the follow up template of the Brain Tumor Reporting and Data System (BT-RADS) (Weinberg et al. 2018). Referring clinicians found reports written with this template less ambiguous than free prose (Gore et al. 2019), and the template reduced hedging (Zhang et al. 2020). Like the template, the report gives the position in treatment, the comparison with the previous scan and the change in enhancing tumour and FLAIR abnormality, here as measured volumes. It leaves out the template’s category and management recommendation, because PRISM offers no diagnosis or advice, and it leaves out steroid and bevacizumab use, which the cohorts do not record (Supplementary Section S24).

2.5. Implementation and computational cost

PRISM is written in Python 3.11 with scikit-learn 1.7.2 (Pedregosa et al. 2011), and its language model is served by vLLM 0.31.0 (Kwon et al. 2023) in 16-bit precision on one NVIDIA H100 GPU with 80 GB of memory. The model weights take 50.2 GiB. The scoring stages are light. On one core of an Intel Xeon Platinum 8481C processor, fitting one of the three models with its penalty search on data of MU’s size (181 scans) took under 0.1 s, scoring took about 2 µs per scan when scans were scored together, and computing the conformal threshold took under 1 ms. Writing reports took 1.08 s per report when generated in a batch, after about two minutes to load the model. These costs leave out segmentation, because we used the masks released with each cohort. A deployment would add the time of an automatic segmentation model. Every stage is one command of a single

package with fixed random seeds, and each run records the hashes of its inputs, code and outputs (Supplementary Section S16).

3. Evaluation design

3.1. Cohorts and the task

Grade 4 glioma is the primary population in all three cohorts (Table 2). The task is to decide whether each scan is the one at which the cohort documented progression. Predicting progression within 120 days of a scan is a secondary task. On MU the intended use is scans from the start of radiotherapy onwards, and MU results are for the intended use unless stated.

3.2. How each cohort was evaluated

Each cohort played one role (Table 3). MU was the development cohort. On it we evaluated every model by five repeats of fivefold cross validation grouped by patient, so that no patient contributed to both fitting and testing, and report the mean pooled held out AUC. We used repeated splits because single splits are unstable in small cohorts (An et al. 2021; Varoquaux 2018). An earlier radiomics model on MU, for example, reached 0.80 on one test split but averaged 0.72 when selected and tested again over 500 splits of the same design (Supplementary Section S8 and Figure S2). We then fitted each model once on all 181 grade 4 detection scans of MU, from every phase of care, locked it, and applied it unchanged to LUMIERE and UCSF under plans committed before any score was compared with an external label. The plans named all MU detection scans, but we fitted on the grade 4 scans, and fitting on all 229 scans of gliomas of every grade changed the LUMIERE AUC by less than 0.01. As a reference, we also fitted each model on UCSF itself by cross validation.

Discrimination is the AUC for telling the scan at which progression was documented from scans without progression. Every AUC carries a 95% confidence interval (CI) from 1,000 bootstrap samples of patients, and on MU these intervals resample held out predictions without the variance from refitting. Conformal sets are judged by coverage, the share of scans whose set contains the true outcome, against a planned threshold of 78% at the 80% level. The agreement statement is judged by whether the combined score ranks scans less well where the two models disagree, and by whether setting those scans aside helps more than removing as many scans matched for phase of care. Reports are judged by the share that pass the checker, by whether every report in a thin situation says so, and by whether reports follow a deliberately changed record.

3.3. Tests fixed in advance

Table 5 lists the questions fixed in advance and their answers. Results of analyses not fixed in advance carry a dagger (†). The disagreement flag and its two tests were planned, but the median rule that defines disagreement was written down after the results had been seen, so every disagreement result carries a dagger.

Table 2. The three cohorts. What each cohort records, and the scans in the detection task. Grade 4 glioma is the primary population.

	MU-Glioma-Post	LUMIERE	UCSF-ALPTDG
Access	Public	Public	Data use agreement
Scans per patient	Up to 6 scans with a label over 120 days (median 2)	Up to 10 scans with a label over 120 days (median 2)	One consecutive pair
Treatment dates	Surgery, radiotherapy start and end, dose, chemotherapy	Days from surgery only; rater flag for scans within three months of radiotherapy	Days from surgery; radiotherapy start only
Documented progression (the label)	Documented date in the clinical record	Expert response rating of each scan	Progression date supplied with the dataset
Segmentation masks	Automatic masks released with the dataset	DeepBraTumIA (three labels)	Masks provided with the dataset
Patients[a]	126	61	140
Age, median	62.5 (IQR 53–70)	62 (IQR 53–70)	57.5
Male	66 (52%)	32 (52%)	81 (58%)
IDH wildtype / mutant / unknown	112 / 8 / 6	38 / 1 / 22	81 / 13 / 46
MGMT methylated / unmethylated / unknown	45 / 62 / 19	26 / 28 / 7	71 / 43 / 26
Detection task[b]: scans with / without progression (patients)	62 / 84 (80) in the intended use; 64 / 117 (92) over all scans	73 / 105 (79)	29 / 40 (69), second scan only
Phase of care in the detection set	Progression: 56 of 62 later after radiotherapy; no progression: 51 of 84 early after radiotherapy	Not recorded	65 of 69 later after radiotherapy[c]

[a] Patient characteristics describe the cohorts of the secondary task over 120 days at MU-Glioma-Post (grade 4) and LUMIERE, and all grade 4 patients at UCSF-ALPTDG. Age is at diagnosis at MU-Glioma-Post and at surgery at LUMIERE.

[b] Positives are the scan at which progression was documented; negatives are defined per cohort in Table 3. The LUMIERE detection set has more patients than the cohort over 120 days because it includes patients who contribute only a progression scan. At MU-Glioma-Post, the 146 scans of the intended use are the 181 grade 4 detection scans less the 35 taken before radiotherapy started, and 42 grade 4 progression scans of patients with no scan carrying a label over 120 days are excluded. Reports were evaluated on the 114 intended use scans of the 60 MU-Glioma-Post patients not used to develop the prompt, on 162 of the 178 LUMIERE scans and on 130 of the 140 grade 4 second scans at UCSF-ALPTDG, the rest belonging to development patients (Section 4.3).

[c] UCSF-ALPTDG gives only the first radiotherapy start; the phase assumes a six week course. Two scans were early after radiotherapy and two had no radiotherapy date.

Table 3. How each cohort was evaluated. Definitions of the label and the procedure in each cohort. Scan counts are in Table 2.

	MU-Glioma-Post	LUMIERE	UCSF-ALPTDG
Role	Development	External test	External test
How models were applied	Five repeats of fivefold cross validation grouped by patient; then fitted once on all grade 4 detection scans and locked	Locked MU models, applied once	Locked MU models, applied once; models refitted on UCSF only as a reference
Scan with progression	First scan on or after the documented progression date	First scan rated as progressive disease	Second scan of the pair, with progression dated within 30 days and closer to it than to the first scan
Scan without progression	Followed by at least 120 days without progression	Earlier scan rated stable or responding	Second scan followed by at least 120 days without progression
Scores evaluated	All scores in Table 1	Clinical data alone, reduced clinical-history, imaging and combined models	Clinical data alone, reduced clinical-history, imaging (both forms) and combined models
Measures	AUC with a patient bootstrap CI; conformal coverage; agreement tests; card tiers; checker pass rate on the 60 patients not used to develop the prompt	AUC with a patient bootstrap CI; conformal coverage; agreement tests; card tiers; checker pass rate	As LUMIERE, with the reports of all grade 4 second scans of patients not used to develop the prompt
Plan committed before the test	Detection plan (2b4c927); system plan (e7cba80)	Detection plan (2b4c927); system plan (e7cba80)	UCSF plan (0e7c9fe); system plan (e7cba80)

Table 4. How each part of the record was validated. Each element PRISM reports for a scan, the evidence that it can be trusted, and where the evidence is reported.

Element of the record	Validated by	Section
Tumour volumes and their change	Taken from the masks released with each cohort. A new segmentation of MU agreed with the released enhancing tumour masks with a median Dice coefficient of 0.80 (Supplementary Section S14); no new expert reading	2.1, 5
Clinical-history, imaging and combined scores	AUC on MU by repeated cross validation, and of locked models at LUMIERE and UCSF	4.1
What carries a score	AUC within phases of care, enhancing volume alone, and the imaging model built step by step	4.4
Conformal set	Coverage overall and by outcome in each cohort	4.2
Agreement of the two models	AUC of the combined score where the models agree and where they disagree; removal matched for phase of care	4.2
Reliability card	Share of scans in tested situations; error by tier	4.2
Report	Checker pass rate in each cohort; thin situations stated; reports following a changed record; seven local models	4.3
Usefulness to clinicians	Not tested; a reader study is designed (Supplementary Section S21.1)	5

Table 5. Tests fixed in advance. Each rule was committed to the repository before the analysis it governs. AUC differences with 95% CIs.

Question	Rule fixed in advance	Outcome
Detection on MU-Glioma-Post		
Do image measurements add to the clinical-history model?	Planned model ≥ 0.75 (lower bound ≥ 0.65); gain ≥ 0.03, CI above 0	**No**: AUC 0.837 (0.763–0.904); gain +0.015 (−0.032 to +0.063)
Do deep networks tuned on these data add?	Gain ≥ 0.03, CI above 0; control with permuted labels covers 0.5	**No**: no network gained; control valid, 0.483 (0.436–0.527)
External cohorts (LUMIERE, UCSF-ALPTDG)		
Combined score at LUMIERE	AUC ≥ 0.65	**Yes** (Section 4.1)
Imaging model at UCSF-ALPTDG	AUC ≥ 0.65, lower bound > 0.5	**Yes** (Section 4.1)
Locked imaging model adds to locked clinical data at UCSF-ALPTDG	Gain, CI above 0	**Yes**: +0.248 (+0.096 to +0.392)
Deep networks learn at UCSF-ALPTDG	Above the control with permuted labels; control covers 0.5	**Yes**: +0.201 (+0.094 to +0.306) over a control of 0.541 (0.480–0.603)
The system (MU-Glioma-Post, LUMIERE, UCSF-ALPTDG)		
Conformal coverage	≥ 0.78 at the 80% level in each cohort	**Yes** for the combined score, on point estimates; not for the imaging model at LUMIERE (0.708) or the clinical-history model at UCSF-ALPTDG (0.696) (Section 4.2)
Does the combined score rank flagged scans less well?	AUC unflagged minus flagged, CI above 0	Disagreement†: **MU only**. Set holding both outcomes: **MU and LUMIERE**[a]
Does setting flagged scans aside beat matched removal?	Retained minus AUC after matched removal, CI above 0	Disagreement†: **MU only**. Set holding both outcomes: **UCSF only**[a]
Do reports pass the checker?	≥ 95% in each cohort	First evaluation **No** (every LUMIERE report failed); after one revision **Yes** (Section 4.3)
Is the tier stated where evidence is thin?	100%	**Yes** (Section 4.3)
Do reports follow a changed record?	≥ 27 of 30 perturbed records	**Yes** (Section 4.3)

Plans: detection (commit 2b4c927), UCSF-ALPTDG (0e7c9fe), the system (e7cba80), second report evaluation (4f7e7b9). Matched removal keeps the mix of care phases in the removed scans (random removal at LUMIERE, which has no phase). Over all scans of gliomas of all grades (a planned secondary population), the planned model gained +0.041 (+0.006 to +0.081) over the clinical-history model, which may reflect its schedule inputs (Section 4.4).

[a] With each cohort's own medians, the rule the reports use; the median rule was written down after the results had been seen†. With medians fixed on MU, disagreement also met both tests at LUMIERE.

3.4. Audits of what carries a score

Each label records how a cohort documented progression, so a score can partly reproduce the way progression was called rather than what the scan shows. On MU, 48 of the 64 scans with progression were taken on the day progression was documented. We used four audits to see what information carries a score. The first is the AUC within phases of care, which pools only pairs of a scan with progression and a scan without progression from the same phase, so that treatment timing cannot separate them. The second is enhancing tumour volume alone, with no fitted model, as a reference for how far a label follows enhancement. The third builds the imaging model in three steps, from the current scan's measurements to change from earlier scans and then three schedule inputs, and compares the AUC across and within phases at each step. The fourth tests whether deep networks see more than the measurements. We fine tuned SwinUNETR segmentation networks to score scans directly and compared them with the same networks trained on permuted labels (Supplementary Section S22). Only the fourth audit was fixed in advance.

4. Results

4.1. The scores ranked scans on MU and, without refitting, in two external cohorts

In the intended use on MU (146 scans), the clinical-history model detected the progression scan with an AUC of 0.82 (95% CI 0.75–0.90) and the imaging model with 0.71 (0.63–0.78)[†]. Clinical data alone scored 0.53 (0.45–0.61; Figure 2). The combined score also reached 0.82 (0.74–0.89)[†], with no detectable gain over the clinical-history model (−0.001, −0.05 to +0.05). A planned model that added image measurements and three schedule inputs to the clinical-history model also fell short of its target gain of 0.03 (Table 5). Within phases of care the clinical-history model's gain was not detectable[†], although its interval cannot exclude a gain as large as the overall one (Section 4.4). Forecasting progression within 120 days was harder, at 0.60 (0.52–0.69) in the intended use (Supplementary Section S23).

The locked combined model, in the reduced form used externally (planned, 0.80, 0.73–0.89, on MU by cross validation), reached 0.77 (0.68–0.86) at LUMIERE and met its planned transfer rule (Table 5). Its gain over the reduced clinical-history model[†] was 0.08 (−0.01 to +0.16). At UCSF both planned imaging models transferred, the measurements from one scan reaching 0.80 (0.69–0.89) and the imaging model 0.73 (0.60–0.84); with coefficients fitted on MU, the imaging model scored 0.07 below the model using one scan (−0.13 to −0.01), and retrained at UCSF it gained 0.14 (0.03–0.26) over its locked version. The reduced clinical-history model scored 0.56 (0.42–0.69) at UCSF, where its radiotherapy flag is approximated. The locked combined score reached 0.86 (0.76–0.93), but part of this reflects scan timing, because the interval between the two scans was shorter before progression[†].

As a probability, the combined score showed no detectable miscalibration in the large at LUMIERE[†] (observed over expected 0.97, 0.81–1.14) but overpredicted progression at UCSF (0.72, 0.55–0.91), and a threshold at the MU median[†] called 90% of UCSF scans positive.

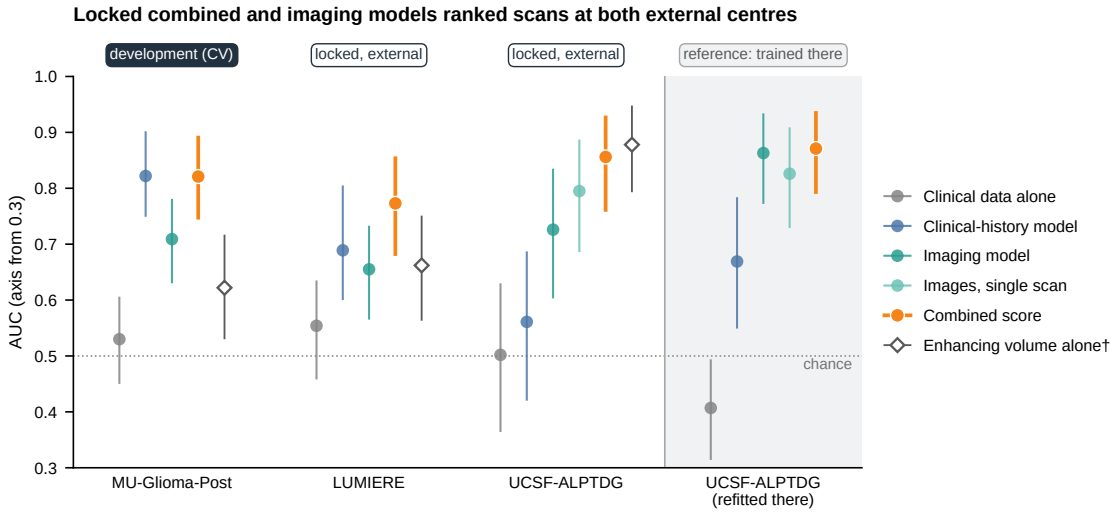


Figure 2. The combined and imaging models rank scans at two external centres without refitting. AUC for detecting the scan at which progression was documented. MU used repeated cross validation in the intended use. LUMIERE and UCSF used models fitted once on MU and applied unchanged, with the reduced clinical-history inputs. The UCSF refit is a reference, where clinical data alone fall below 0.5, consistent with the downward bias of AUC estimated by cross validation without signal. At MU the combined score is its full form. Externally, it has a reduced form. At UCSF the images are shown for the model using one scan and for the imaging model. The imaging model and combined score on MU and the clinical data only, reduced clinical-history and imaging models at LUMIERE were not fixed in advance†. Open diamond: log enhancing tumour volume alone† (at LUMIERE, the 163 scans with enhancing tumour). Points have 95% CIs from bootstrap samples of patients on an axis from 0.3. The dotted line marks chance.

4.2. The conformal sets kept their coverage on point estimates, and disagreement marked weaker ranking at MU

The 80% conformal sets of the combined score met the planned coverage threshold of 78% on point estimates in all three cohorts, covering 78% to 87% of scans (Figure 3A). The intervals at MU (76% to 88%) and UCSF (68% to 87%) reached below the threshold, and the imaging model's set at LUMIERE and the clinical-history model's set at UCSF fell below it. Coverage differed by outcome at the external centres†. LUMIERE covered 77% (67% to 87%) of progression scans and 94% of scans without progression, and UCSF 97% of progression and 65% (51% to 79%) of scans without progression. Calibrating the sets separately for each outcome on MU did not fix this, and coverage for scans without progression fell to 73% (63% to 82%) even on MU.

Disagreement between the clinical-history and imaging models† marked scans that the combined score ranked less well at MU, but not at the external centres. On MU the combined score's AUC was 0.31 higher where the two models agreed (0.18–0.43), and setting the disagreeing scans aside gained 0.08 (0.05–0.12) over removing as many scans matched for phase of care, meeting both planned tests (Table 5; Figure 3B). With each cohort's own medians, the rule the reports use, neither test was met at LUMIERE or UCSF. With medians fixed on MU, both were met at LUMIERE but not at UCSF (Figure 3B). A conformal set holding both outcomes, used as a flag, met one of the two planned tests in each cohort (Table 5).

Only four situations on the reliability card† reached the tested tier (Figure 3C), one at MU and three at LUMIERE, where no phase of care is recorded and situations differ by enhancement only.

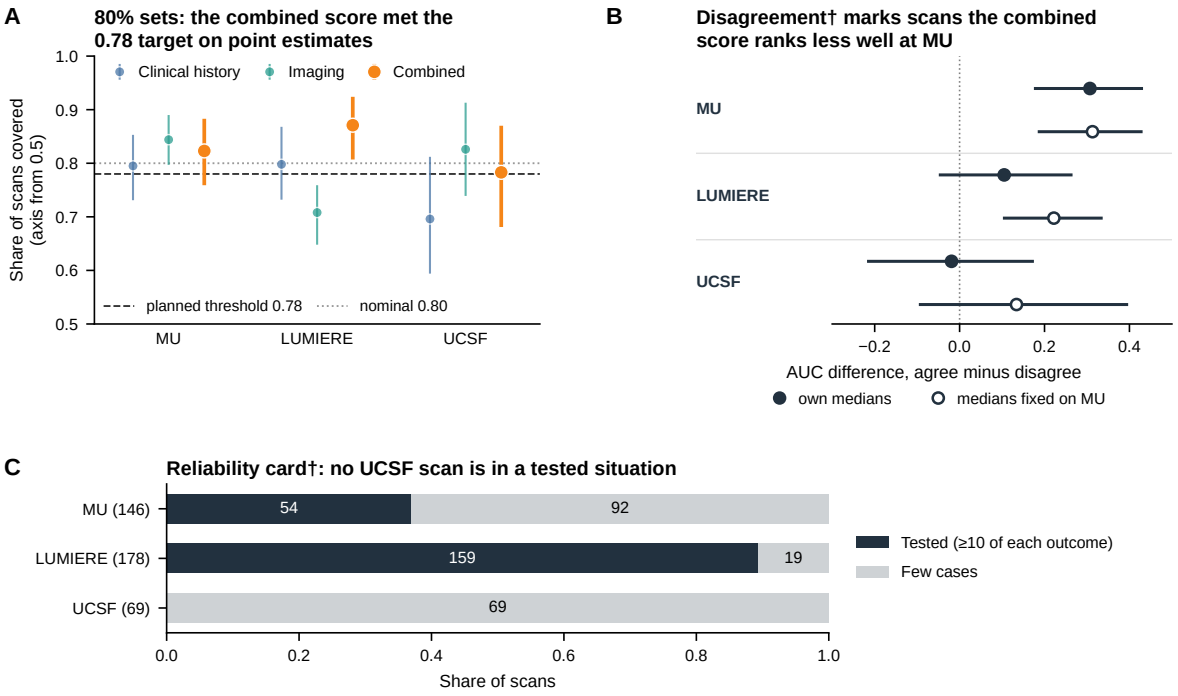


Figure 3. The combined score's conformal sets meet their coverage target on point estimates. Disagreement† marks scans where the combined score ranks less well at MU. (A) Coverage of the 80% conformal sets by cohort and model. The dashed line marks the planned threshold (0.78), and the dotted line marks the nominal level (0.80). (B) The combined score's AUC where the two models agree minus where they disagree, with each cohort's own medians (filled) and with medians fixed on MU (open, using the other folds' medians on MU). The tests were planned, but the median rules were not†. (C) Scans whose situation reached the reliability card's† tested tier. The rest had few cases, and none was untested.

No scan in these sets was in an untested situation. Every UCSF situation had few cases, and at MU no situation in the first 12 weeks after radiotherapy held more than three progression scans. In the planned descriptive comparison, tested situations had lower error than situations with few cases at LUMIERE (29% against 42% of scans misclassified at the cohort median) but not at MU (29% against 24%), where the scans with few cases are mostly clearly without progression early after radiotherapy.

For a large share of scans, these statements tell the reader something a score alone does not. The conformal set held both outcomes for 25% of scans at MU, 38% at LUMIERE and 36% at UCSF. The two models disagreed on 40%, 57% and 55% of scans†, and 63%, 11% and 100% of scans fell in situations with few cases†. Where the set held both outcomes, the combined score misclassified 37%, 48% and 32% of scans when every scan above the cohort's median score was called progression, against 22%, 20% and 25% of the other scans†.

4.3. After one revision of the prompt, every report agreed with its record

Every report written by Qwen3.8-27B passed the checker, 114 at MU, 162 at LUMIERE and 130 at UCSF. Of six other local models given the same prompt, which had been revised for Qwen3.8-27B, only Gemma-4 31B also did (Figure 4B). All 220 reports in situations with few cases or untested situations said so (71 at MU, 19 at LUMIERE and 130 at UCSF). When one confidence or measurement field was changed in 30 records from the three cohorts, 28 reports followed the change, against a planned minimum of 27 (Table 5). These results followed one revision of the prompt†. Before it, every LUMIERE report failed because a fixed sentence quoted a number held in no field, so we corrected the prompt, left the checker unchanged and evaluated again on patients not used for the revision. Whether the segmentation itself is right lies outside what the checker can see.

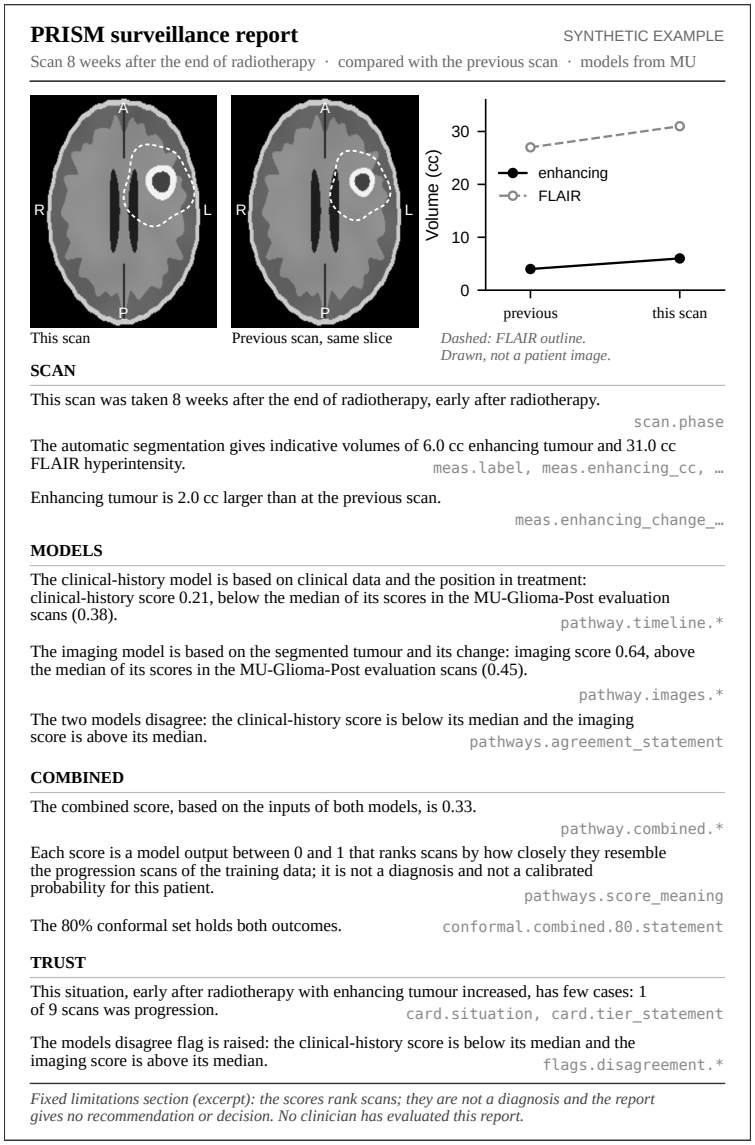
4.4. The scores rested on the treatment schedule at MU and on enhancement at UCSF

At MU the clinical-history and combined scores depended mainly on where the patient was in treatment, and at UCSF the imaging scores followed the volume of enhancing tissue† (Figure 5). At MU most progression scans came more than 12 weeks after radiotherapy and most scans without progression within those 12 weeks (Table 2), so the schedule alone could separate most of them. Most pairs within a phase come from the later phase (56 progression and 32 scans without progression). Adding treatment history to clinical data and images† changed the AUC by +0.11 (0.04–0.20) across phases of care and by −0.02 (−0.13 to +0.11) within them (Figure 5A). Within a phase, the imaging model scored 0.68 (0.58–0.77) and the clinical-history model 0.64 (0.51–0.78), with no difference detectable.

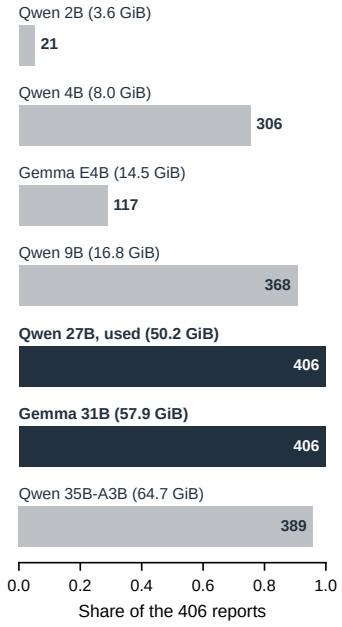
On MU the imaging model gained most from inputs that carry timing†. Adding change from earlier scans to the current scan's measurements, which gives the imaging model, raised the AUC by 0.05, with an interval that included zero (−0.003 to +0.106). Adding the planned model's three schedule inputs (Table 1) raised it by a further 0.10 (0.05–0.15) across phases of care but not within them (−0.01, −0.06 to +0.04) (Figure 5C).

At UCSF the progression labels coincided with a rated increase in enhancement†, so a model could score well there by tracking it. UCSF rated enhancement as increased at 27 of the 29 progres-

A **A PRISM report (synthetic example)**



B **Reports passing the check**



Dark: ≥ 95% in every cohort; light: below 95% in at least one. Same prompt for all.

Figure 4. What the clinician receives, and the check. (A) A report for a synthetic scan eight weeks after radiotherapy with growing enhancement (Section 5). The segmentation figure is drawn here rather than taken from a patient, and the narrative is synthetic. Volumes and scores are invented, and the card row is MU's. Each sentence lists, in grey, the record fields it cites. The delivered report also holds tables of every value and a fixed limitations section (excerpt shown). (B) Reports passing the check, of 406, for seven local language models given the same prompt (Qwen3.5 2B, 4B and 9B, Gemma-4 E4B and 31B, Qwen3.8-27B used in PRISM, and Qwen3.5-35B-A3B), after one revision of the prompt† (Section 4.3). Dark bars show models for which at least 95% passed in every cohort, and the model weight size is in brackets.

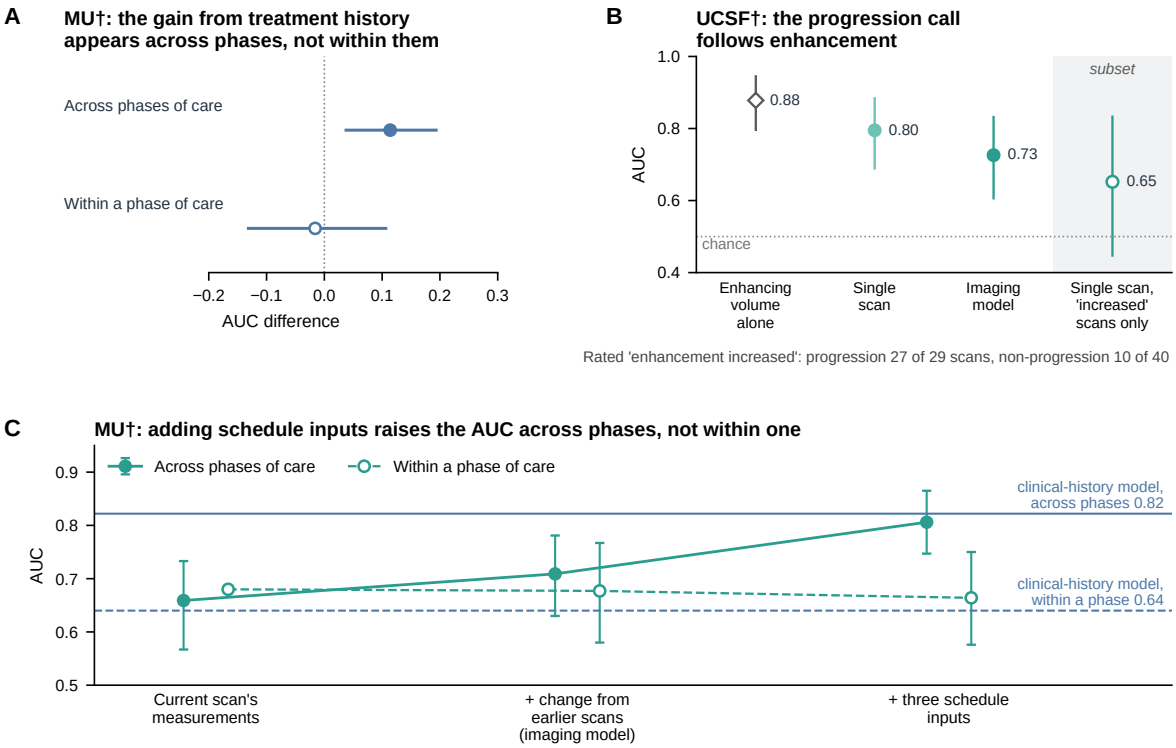


Figure 5. The scores followed the schedule at MU and enhancement at UCSF. (A) MU: gain from adding treatment history to clinical data and images, across and within phases of care. (B) UCSF: enhancing volume alone beside the locked imaging models; the open point is restricted to the 37 scans rated as showing increased enhancement. (C) MU: the imaging model built in three steps, across and within phases of care, against the clinical-history model; the third step adds the planned model's schedule inputs to the image inputs, without its clinical-history inputs. The gain at the second step has an interval that includes zero, and the first point within a phase has no interval. The audits were not fixed in advance (†); the locked imaging models' AUCs over all 69 scans in (B) were planned. Points with 95% CIs from bootstrap samples of patients.

sion scans, and progression was dated a median of 7 days after the scan (Figure 5B). Enhancing volume alone reached 0.88 (0.79–0.95) there, above the point estimates of the locked imaging models, and above its AUC at MU and LUMIERE (Figure 2). Among the 37 scans rated as increased, the model using one scan reached 0.65 (0.44–0.84), so this sample cannot show whether the measurements add to that rating. Deep networks tuned on UCSF scans beat the control with permuted labels (Table 5), and their scores tracked enhancing volume[†].

5. Discussion

PRISM's combined and imaging models ranked scans at two other centres without refitting, but the evidence behind a score depended on how each cohort was scanned and labelled. At MU, the score mostly followed the treatment schedule and at UCSF it followed the enhancement on which progression was called[†]. An AUC cannot show which evidence a model used. The audits of Section 4.4 suggested it, and the separate models show the clinician which kind of evidence drives each score. Process variables can carry much of the apparent skill of an image model (Badgeley et al. 2019), and image models for glioblastoma surveillance have been benchmarked separately at the first and second scans after radiotherapy (Guo and Mirzaei 2025); we found no earlier study that compared image and treatment history evidence within phases of care, or showed that what carries the same locked score differs between centres tested externally. The risk scale did not transfer to UCSF, so a new centre would recalibrate the score and set any threshold on local scans.

Because its modules are separate, PRISM runs on what a cohort holds, given scans and masks. The imaging model needs only those, which most public datasets have, and a version using one scan serves where there is no earlier scan. The clinical-history model is added where dates exist. The reliability card is a lookup table, so a new cohort adds rows to it and a situation's tier can change as evidence accumulates.

Consider a scan eight weeks after radiotherapy in which the enhancing tumour has grown (Figure 4A). The clinical-history model scores such a scan low, because most scans at this point in treatment are not progression, and the imaging model scores it high, because enhancement has grown. The report shows both, states that they disagree and that the conformal set holds both outcomes, and records from the card that the situation has few cases (at MU, 1 of 9 such scans was progression[†]). The evidence PRISM holds for this situation is therefore too thin to support a call, in the weeks when pseudoprogression is most common, and the clinician decides with information it does not use, such as symptoms, steroid dose and the radiotherapy plan.

Each component of PRISM has been described before: longitudinal segmentation with measurement of change (Rudie et al. 2022), late fusion of separately trained image and clinical models (Boehm et al. 2022), fusion of clinical and imaging data to separate pseudoprogression from progression (Gomaa et al. 2025), conformal prediction for glioma segmentation (Elyassirad et al. 2025) and for triage at a deployment site (Angelopoulos et al. 2024), disagreement between modalities as uncertainty (Mahapatra et al. 2026), reliability from performance on similar cases (Habas et al. 2007), model cards (Mitchell et al. 2019), and reports written by local language models from brain tumour measurements (Heras Rivera et al. 2026; Valerio et al. 2025). The nearest systems combine some of these. One assigns a BT-RADS category at follow up from tumour volumes and treatment variables extracted by a language model, at one centre and without an uncertainty measure for each scan (Jabal et al. 2026); another keeps a ledger of evidence with a confidence for each claim in lon-

gitudinal neuro oncology, but works from reports rather than images and checks its evidence with a language model (Liu et al. 2026). On these cohorts, earlier work predicted a label for each patient on MU (Christodoulou et al. 2026) and response categories from longitudinal MRI (Tikhonov et al. 2025), and a model trained on MU was applied unchanged at UCSF to predict survival (Ding et al. 2026); separately, Chen et al. (2026) predicted recurrence risk from preoperative and postoperative MRI across centres. None reports image and treatment history evidence separately at each scan, or tests a progression call for each scan locked at two external centres. PRISM therefore combines known parts, showing image and treatment history evidence at each scan with a confidence from measured performance, and tests the combination at two external centres under rules fixed in advance.

PRISM was developed at one centre, and the two external tests are its only evidence that the ranking transfers. The label is progression as each centre documented it; no scan was read again, and at UCSF most progression scans were also rated as showing increased enhancement. Of the confidence measures, only average coverage met its planned threshold at all three centres, and only on point estimates. Disagreement† marked scans the combined score ranked less well at MU alone; that rule needs a batch of a cohort's scans, and the version using one scan (MU medians) was added after the results had been seen. No UCSF scan fell in a tested situation. Subgroups are small, the analyses marked † need testing on new data, and no clinician has yet read the reports. PRISM forecast progression within 120 days less well than it detected progression at the scan (AUC 0.60, 0.52–0.69, in the intended use).

The next steps are to pool the rare situations (early progression after radiotherapy, and growing enhancement without progression) across cohorts, including the Burdenko and UCSD-PTGBM datasets, and test models tuned on them there; and a reader study to show whether separate evidence and a stated confidence change clinicians' decisions.

Conclusion

PRISM keeps evidence from images and treatment history apart and shows the clinician both scores at every scan with how far each has been tested there. Its combined and imaging models ranked scans at two other centres without refitting, while the scores followed the treatment schedule at MU and enhancement at UCSF, a difference the separate models make visible.

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[PENDING AUTHOR DECISION].

Disclosure statement

[PENDING AUTHOR DECISION: competing interests, or "The authors report there are no competing interests to declare."] If any author serves on the editorial or advisory board of this journal, state it here.

Declaration of generative AI use

[DRAFT, PENDING AUTHOR CONFIRMATION:] Generative AI is part of the system studied. PRISM's report generator uses a local language model (Qwen3.8-27B; six other models with open weights in a comparison); all ran offline on the authors' computing cluster and saw only derived system records. An AI assistant (Claude, Anthropic) was used to write and run analysis code and to draft and edit the manuscript under the authors' direction; it was given no data at the patient level through any hosted service. The authors reviewed all code, results and text and take full responsibility for the content of this article.

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Ethics statement

[PENDING AUTHOR DECISION: secondary analysis of datasets with identifiers removed (MU-Glioma-Post and LUMIERE, public; UCSF-ALPTDG under a data use agreement); state the institutional determination or waiver and cite the source datasets' approvals.] No registered protocol. Analysis plans were committed to the repository before each external test; their commits are listed in Table 5. Patient and public involvement: none.

Data availability statement

[PENDING AUTHOR DECISION: MU-Glioma-Post (TCIA, doi:10.7937/7K9K-3C83) and LUMIERE (figshare; Suter et al. 2022) are public; UCSF-ALPTDG is available from its provider under a data use agreement; state which derived tables, fold assignments and locked model files are released, with a DOI.] Code: [PENDING: public repository link and release tag.]

Supplemental material

Supplemental data for this article are provided as a separate file.

Notes on contributors

[PENDING: a short biographical note for each author (no more than 200 words each).]

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